Summary of Regression Modeling Concepts

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```r
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
        # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

1 We use different probability models for different data types

- Binary outcomes: Bernoulli[1] models
- Event rate[2] outcomes: Poisson[3]/Negative binomial[4] models
- Time-to-event outcomes: Survival models
- Catch-all: Gaussian[5] models

2 We use different link functions to connect these models with covariates

- Bernoulli models: logit[6] link
- Count models: log link + offset[7]
- Survival models: log link
- Gaussian models: identity link

Figure 1 shows how the various models we have studied have analogous structures: each row uses model-specific transformations to connect the modeled quantity to a linear predictor[8] $\eta(\tilde{x}) = \tilde{x} \cdot \tilde{\beta} + \eta_0$.

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[1] https://morrison-lab.github.io/pds/random-variables.html#def-bernoulli

[2] https://morrison-lab.github.io/pds/distributions.html#def-event-rate

[3] https://morrison-lab.github.io/pds/distributions.html#def-poisson

[4] https://morrison-lab.github.io/pds/distributions.html#def-nb

[5] https://morrison-lab.github.io/pds/random-variables.html#def-normal

[6] binary-outcome-associations.qmd#def-logit-fn

[7] https://morrison-lab.github.io/pds/distributions.html#def-offset

[8] Intro-to-GLMs.qmd#def-linear-predictor

Linear[9] (Gaussian):

$$\underbrace{\mu}_{\text{mean}} \;\underset{\xleftarrow{=}}{\xrightarrow{=}}\; \underbrace{\eta(\tilde{x})}_{\substack{\text{linear}\\ \text{predictor}}} = \tilde{x} \cdot \tilde{\beta} + \mu(0)$$

Logistic[10] (Binomial):

$$\underbrace{\mu}_{\text{mean}} \;\underset{\xleftarrow[\pi \cdot n]{}}{\xrightarrow{\frac{\mu}{n}}}\; \underbrace{\pi}_{\text{probability}} \;\underset{\xleftarrow[\frac{\omega}{1+\omega}]{}}{\xrightarrow{\frac{\pi}{1-\pi}}}\; \underbrace{\omega}_{\text{odds}} \;\underset{\xleftarrow[\exp\{\eta\}]{}}{\xrightarrow{\log\{\omega\}}}\; \underbrace{\eta}_{\text{log-odds}} = \tilde{x} \cdot \tilde{\beta} + \eta(0)$$

Count (Poisson[11]):

$$\underbrace{\mu}_{\text{mean}} \;\underset{\xleftarrow[\lambda \cdot t]{}}{\xrightarrow{\mu/t}}\; \underbrace{\lambda}_{\text{rate}} \;\underset{\xleftarrow[\exp\{\eta\}]{}}{\xrightarrow{\log\{\lambda\}}}\; \underbrace{\eta}_{\text{log-rate}} = \tilde{x} \cdot \tilde{\beta} + \log\{\lambda(0)\}$$

Survival (Cox PH[12]):

$$\underbrace{\mu}_{\text{mean}} \;\xleftarrow[\int_{t=0}^{\infty} \mathrm{S}(t|\tilde{x})dt]{}\; \underbrace{\mathrm{S}(t|\tilde{x})}_{\text{survival}} \;\underset{\xleftarrow[\exp\{-\Lambda(t|\tilde{x})\}]{}}{\xrightarrow{-\log\{\mathrm{S}(t|\tilde{x})\}}}\; \underbrace{\Lambda(t|\tilde{x})}_{\text{cumulative hazard}} \;\underset{\xleftarrow[\int_0^t \lambda(u|\tilde{x})\,du]{}}{\xrightarrow{\Lambda'(t|\tilde{x})}}\; \underbrace{\lambda(t|\tilde{x})}_{\text{hazard}} \;\underset{\xleftarrow[\exp\{\eta(t|\tilde{x})\}]{}}{\xrightarrow{\log\{\lambda(t|\tilde{x})\}}}\; \underbrace{\eta(t|\tilde{x})}_{\text{log-hazard}} = \tilde{x} \cdot \tilde{\beta} + \eta_0(t)$$

Figure 1: Parallel structures of GLMs[13] and survival models. Each row shows the transformation chain from the modeled quantity to the linear predictor $\eta(\tilde{x}) = \tilde{x} \cdot \tilde{\beta} + \eta_0$, where η_0 denotes the model-specific intercept term, and $\eta_0(t)$ denotes the Cox PH baseline log-hazard term.

3 We use maximum likelihood estimation to fit models to data

- likelihood[14]
- log-likelihood[15]
- score function[16]
- hessian[17]

4 We use asymptotic normality of MLEs to quantify uncertainty about models

- observed information matrix[18]
- expected information matrix[19]
- standard error[20]
- confidence intervals[21]
- p-values[22]

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[14] https://morrison-lab.github.io/sds/intro-MLEs.html#def-lik
[15] https://morrison-lab.github.io/sds/intro-MLEs.html#def-loglik
[16] https://morrison-lab.github.io/sds/intro-MLEs.html#def-score
[17] https://morrison-lab.github.io/mds/vector-calculus.html#def-hessian
[18] https://morrison-lab.github.io/sds/intro-MLEs.html#def-oinf
[19] https://morrison-lab.github.io/sds/intro-MLEs.html#def-einf
[20] https://morrison-lab.github.io/sds/estimation.html#def-SE
[21] https://morrison-lab.github.io/sds/inference.html#def-confidence-interval
[22] https://morrison-lab.github.io/sds/inference.html#def-p-value

5 We use (log) likelihood ratios to compare models

Sometimes we adjust these comparisons for model size (AIC[23], BIC[24])

References

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[23] Linear-models-overview.qmd#def-aic

[24] Linear-models-overview.qmd#def-bic